

Molly Cliff

Ph.D. Candidate

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Infectious disease epidemiologist working at the intersection of human health and the environment. Currently completing a PhD at the University of Cambridge investigating the impact of climate change on the distribution and incidence of bacterial meningitis across Africa. Previously worked as an Epidemiology and Information Analyst with the UK Health Security Agency Field Service across a wide range of communicable disease outbreaks, including a secondment to the World Health Organisation as a field epidemiologist on the European MPox response.

Education

Ph.D. Veterinary Medicine

April 2024 - Present

University of Cambridge *Darwin College, Cambridge, UK*

Research Project: Spatiotemporal climatic determinants of bacterial meningitis outbreaks across Africa.

Anticipated graduation: April 2027

Masters of Science Tropical Disease Biology

September 2019 - September 2020

Liverpool School of Tropical Medicine *Liverpool, UK*

Research Project: Geospatial analysis of travel distance from health facilities in Malawi, in areas where acute African trypanosomiasis continues to be a public health issue

Bachelors of Science Biomedical Science

September 2015 - June 2018

University of Kent *Canterbury, UK*

Research Project: What is the correlation between waterborne parasites and flooding

Publications

Published

Cliff, M., Jahn, S., Fouda, A., Latt, A., Lingani, C., & Trotter, C. (2026). Re-evaluating environmental associations with meningitis risk across Africa. *Wellcome Open Research*, <https://wellcomeopenresearch.org/articles/10-497>

Cliff, M., Diallo, K., Yapi, R., Tano, A., Jahn, S., Wainwright, C., & Trotter, C. (2026) How do climatic factors directly influence the incidence and risk of meningococcal meningitis across the African meningitis belt? A narrative literature review. *GeoHealth*, <https://doi.org/10.1029/2025GH001749>

Willgert, K., Cliff, M., Meinke, S., Messina, D., Broom, D. M., Wood, J., Conlan, A. J. K., & Joshua, K. (2025). Burden of Bovine Tuberculosis on Animal Health, Welfare and Production: A Systematic Review. *Transboundary and Emerging Diseases*, <https://doi.org/10.1155/tbed/6541298>

Cliff, M., Welaga, P., Mohammed, N., Ansah, P., Heyderman, R. S., Trotter, C., & Kwambana-Adams, B. (2024). Strategies for controlling pneumococcal disease and outbreaks during humanitarian emergencies. *Nature Medicine*, <https://doi.org/10.1038/s41591-024-02922-x>

In Preparation

Delpont, D., **Cliff, M.**, Trotter, C., Karachaliou-Prasinou, A., Abeysuriya, R., & Scott, N. Impact and efficiency of preventative and reactive vaccination strategies in the African Meningitis Belt: a modelling study

Cliff, M., Perez-Duque, M., Saljie, H., Fouda, A., Latt, A., Lingani, C., Gayawan, E., & Trotter, C. Bayesian Spatiotemporal Modelling of Climatic Drivers of Bacterial Meningitis Outbreaks Across the African Continent

Awards

Charles Slater Travel Award (£2200)

August 2026

School of Biological Sciences, University of Cambridge

Wrote and secured travel grant independently

Research Experience

PhD Researcher

April 2024 - present

Department of Veterinary Medicine, University of Cambridge

Undertaking PhD investigating the impact of climate change on bacterial meningitis distribution and incidence across Africa. Developed a logistic regression model using k-means clustering of temporal climate variables, updating existing continental risk maps. Extended this framework with R-INLA to quantify spatiotemporal climate–disease relationships at regional and continental scales. Currently using bias corrected Global Climatic Model ensembles to project future outbreak risk under several Shared Socioeconomic Pathway scenarios.

Research Assistant

January 2023–March 2024

Department of Veterinary Medicine, University of Cambridge

Conducted epidemiological analysis of pneumococcal meningitis in displaced populations across Africa. Independently examined the relationship between displacement and pneumococcal disease outbreaks, highlighted current data gaps and suggested new research strategies to understand the disease burden within displaced populations.

Research Skills

Statistical modelling: Stata, SPSS, R (R-INLA, sf, terra), spatiotemporal modelling, Bayesian inference, Large dataset analysis, Global climatic models

Climatic data: NetCDF/raster data processing, ERA5, CHIRPS, CMIP6 climate scenarios

Reproducibility: Git/GitHub, RMarkdown

Professional Experience

Epidemiology and Information Analyst

October 2020 - January 2023

UK Health Security Agency, Cambridge, UK

Led field response to a complex MRSA outbreak under significant time pressure: generated a case definition, analysed whole-genome sequencing data, and produced a formal outbreak report. Independently designed the contact tracing data collection template used by hospital and ambulance

staff during a Crimean-Congo Haemorrhagic Fever incident, building an organised and adaptable system under rapidly evolving conditions. Conducted a logistic regression analysis of risk factors for late HIV diagnosis in the East of England. Developed and maintained the care home COVID-19 surveillance report, producing outputs to diverse stakeholder specifications using R and RMarkdown. Piloted a syphilis prevalence survey across GUM clinics in the region using Select Survey, providing an overview of patient symptoms and treatment patterns.

GOARN Epidemiologist

August 2022 - September 2022

World Health Organisation European Region- Copenhagen, Denmark (UK Remote)

Conducted R coding updates to the joint WHO/ECDC epidemiological bulletin, incorporating new analyses in response to evolving data and stakeholder requests across affected European countries. Co-produced an RMarkdown version of the WHO/ECDC bulletin in Russian using the Cyrillic alphabet, working with translators to extend reach to Russian-speaking public health audiences. Contributed to the design of a cross-national survey examining surveillance and response measures implemented across European countries, supporting real-time monitoring of the outbreak response.

Teaching Experience

Foundations of Evidence Based Practise Module

September 2024 - present

Seminar Leader/Supervisor *Faculty of Biology, University of Cambridge*

Carried out small-group teaching on introductory biostatistics and epidemiology across three groups of 4–6 first-year Medical and Veterinary students. Additionally led critical analysis of several academic papers.

Presentations

External conferences

International Pathogenic Neisseria Conference

September 2026

Busselton, Western Australia

Flash Talk and Poster: Bayesian Spatiotemporal Modelling of Climatic Drivers of Meningitis Outbreaks Across the African Continent

Oral Presentation: Risk-based Campaign Strategies for the African Meningitis Belt

Presented on behalf of Prof. Caroline Trotter, University of Cambridge / Gates Foundation

Infectious Disease Dynamics Conference

July 2026

Ambleside, UK

Poster: Bayesian Spatiotemporal Modelling of Climatic Drivers of Meningitis Outbreaks Across the African Continent

International Pathogenic Neisseria Conference

April 2025

Florence, Italy

Poster: Associations between environmental factors and epidemic meningitis risk across Africa

Invited Presentations

Cambridge Zero Symposium

November 2025

University of Cambridge, Cambridge, UK

Invited Speaker: How do climatic factors influence outbreaks of meningococcal meningitis across the African meningitis belt?

Meningitis Research Foundation Conference

November 2025

London, UK

Invited Speaker: Associations between environmental factors and epidemic meningitis risk across Africa

Voluntary Experience

Writer

March 2021 - January 2023

DYYV Global, Cambridge, UK

Wrote monthly articles on global public health and tropical infectious disease epidemiology; conducted interviews with global health charities on disease elimination campaigns.

Research Assistant

December 2019

Liverpool School of Tropical Medicine, Liverpool UK

Assisted with systematic review on HIV interventions and mental health outcomes in LMICs.

Youth Steering Group Member

May- August 2019

Voluntary Services Overseas, London, UK

Co-organised the ICS Youth Employability Event (60 attendees); co-moderated the discussion panel.

Public Health and Epidemiology Intern

January-May 2019

Lepra, Colchester, UK

Analysed active case-finding data for leprosy elimination in India; conducted interview for International Women's Day with a leading health rights activist.

Volunteer

July 2018- January 2019

Global Alliance for Rabies Control, Cambridge, UK

Worked to put together a discussion document for internal staff at GARC regarding the WASH programme within GARC and potentially collaborations with similar organisations.

International Citizen Service Volunteer

October- December 2018

Voluntary Services Overseas, Mongla, Bangladesh

Delivered sexual and reproductive health training to community educators and students as part of a multidisciplinary field team.

Professional Activities

- Peer reviewer: *PLOS ONE*; *Meteorological Applications*
- Member: *Vaccine Impact Modelling Consortium (2024-Present)*; *The John Snow Society (2025-Present)*